PROJECT AND TEAM INFORMATION

Project Title

Smart Traffic and Emergency Route Management

Student / Team Information

Team Name: *Team #T079*	
Team member 1 (Team Lead) *Bhimani,Nishil : 24021985: 24021985@geu.ac.in*	*Bhimani,Nishil – 24021985 24021985@geu.ac.in* 
Team member 2 *Gupta, Aastik: 240211350: 240211350@geu.ac.in*	*Gupta, Aastik – 240211350 240211350@geu.ac.in* 
Team member 3 *Mangukiya, Jasmin: 240211744: 240211744@geu.ac.in*	*Mangukiya, Jasmin – 240211744 240211744@geu.ac.in* 
Team member 4 *Agarwal, Rishi: 240212395: 240212395@geu.ac.in*	*Agarwal, Rishi – 240212395 240212395@geu.ac.in* 

PROPOSAL DESCRIPTION

Motivation

Urban traffic congestion has become one of the most important problems that have been occurring in todays world. It can cause a lot of delay to people, increasing fuel consumption and pollution, but more importantly, it can obstruct the paths of emergency vehicles. It is crucial to provide emergency services with the fastest possible way to reach the scene of the call. and multiple medical studies have shown that a patients chance to survive an emergency significantly depends on the time between the moment of emergency call and the moment of arrival of emergency vehicles. but most traffic light systems cannot currently react to an emergency call. They can only regulate the traffic flow on a specific set of rules pre set in it or make it easier for moving cars to pass through the intersection, which means nothing to other types of vehicles. thats why emergency vehicles are forced to stop and wait for when it will be their turn to go, despite being in a hurry to arrive at the scene of emergency.I believe that there is a need to create a system that will allow emergency vehicles to reach the scene of emergency faster by analyzing the current state of traffic lights and finding the fastest way to it. Then the system would have to find a way to show that emergency vehicle to other cars on the road and change the traffic lights timing in order to form a path for the emergency vehicle.our project is concerned with developing such a system. By analyzing the traffic data, I hope to find patterns that will allow me to calculate an approximate time that would take for an emergency vehicle to get to the scene of emergency under specific conditions. Then we'll have to apply the changes to the traffic light system. The changes would allow emergency vehicles to ignore the traffic lights and proceed to the scene of emergency without risking any other cars . As a result, it'll be able to show how my system will reduce the travel time of emergency vehicles and as a result increase the chances of a patient surviving an emergency. and we believe that the project can also help to reduce the overall traffic congestion, which will benefit other cars passing through the city. at last working on this project will give me an opportunity to work with modern technologies and learn about current traffic regulations and standards. If the project is successful it might also become useful for a real-world traffic light system in one of the cities, benefiting its traffic congestion and helping it to reduce pollution and optimize traffic light scheduling..

State of the Art / Current solution

Most of the urban traffic signal systems today are pre set programs which do not always react to the road conditions. Some use loop sensors or even cameras to register the passing cars and adjust the timing of the traffic signals accordingly, but even those neglect emergency vehicles. There are commercial solutions, such as in Israel, where the traffic light priority system lets ambulances, police cars, and fire trucks pass with green lights on the way to an emergency, and Texus-based Opticom utilizes GPS and infrared sensors for a similar purpose. They are expensive and require large-scale infrastructure investments to be installed in each traffic light. Several countries have developed computer algorithms for emergency vehicles routing, which take into consideration the shortest possible distance and the degree of traffic, but they utilize sparsely connected CAD systems and only make recommendations for the fastest route based on the estimated time of arrival. The navigation systems for civilians, such as Google Maps or Waze, do not have special traffic light handling algorithms for emergency vehicles. Most of the emergency services in developing countries still rely on driver's reaction to sirens and flashing lights.A system which could calculate the optimal route and adjust the traffic signals to let the emergency vehicle pass with minimal stops without requiring expensive hardware investments and without being tied to a particular city's infrastructure would be extremely useful.

Project Goals and Milestones

Goals

- *Design a system which finds the fastest route for an emergency vehicle according to traffic conditions*
- *Allow for dynamic communication with traffic lights to cut down on stops for emergency vehicles*
- *Create a dashboard for traffic authorities and dispatchers to monitor emergency vehicle's progress and traffic conditions*
- *Notify drivers around the emergency vehicle to allow for their presence and attempt to clear the road a reduction in test response time over a predetermined baseline.*

Milestones

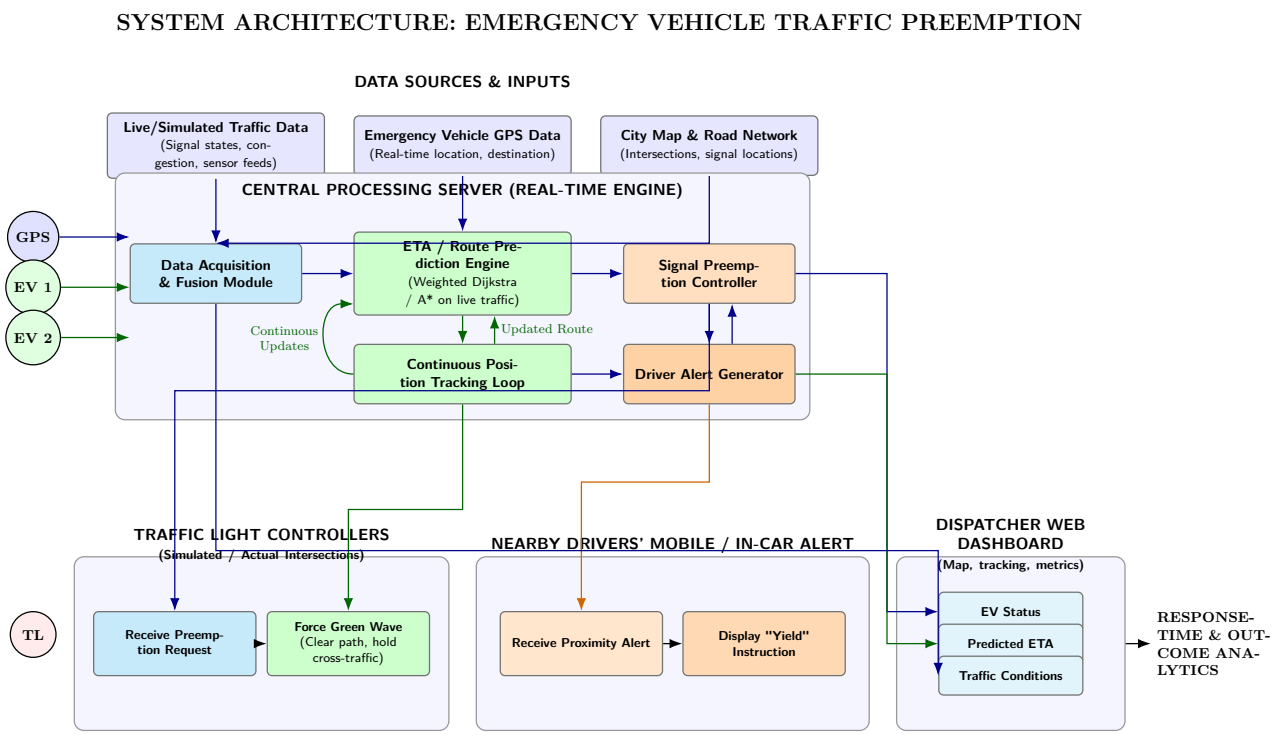
1. *Research and Requirement Analysis.*
2. *Data Layer Setup and Traffic Integration.*
3. *Route Optimization Engine Development.*
4. *Signal Coordination Module Implementation.*
5. *Dashboard and Driver Alert Development.*
6. *System Integration and Testing.*
7. *Final Evaluation, Documentation, and Demonstration.*

Project Approach

The system is designed with a modular real time architecture. There are three main modules: data acquisition, route optimization, and alert/signal coordination.
Data Acquisition: *Traffic data can be acquired from Google Maps real time traffic API or similar open-source platforms. It should be used in combination with data from the GPS devices of emergency vehicles.*
Route Optimization: *A graph based routing algorithm is used for finding the shortest path from the ambulances current location to the destination. Such algorithms as Dijkstras or A with dynamic weighting can be used. The weights of the graph edges are adjusted depending on the traffic situation.*
Signal and Alert Coordination: *2) when the ambulance approaches the intersection, the control module requests signal priority to display a green light. At the same time drivers nearby may receive a push notification via a companion app or dashboard.*
System Coordination:*Overall modules interact with each other through a backend server that stores information about the state of vehicles and the road. The dashboard displays information about the ambulance location and route. The backend can be implemented in real-time databases such as Firebase or MongoDB.*

System Architecture (High Level Diagram)

article [a3paper, landscape, margin=1cm]geometry tikz graphicx



The project will produce a working prototype of the Smart Traffic & Emergency Route Management system which will include:

- *A route optimization engine which will find optimal congestion aware routes for emergency vehicles*
- *A simulation of signal preemption to reduce time spent waiting at crossings*
- *A web dashboard for dispatchers to monitor the progress of their units as well as the current traffic conditions*
- *An in-car driver notification system which will warn drivers of an approaching emergency vehicle*

The project will also produce a comparison report of the response times of systems using optimized routing versus systems without it. Other deliverables include the full working prototype, source code, documentation, a video demonstrating the project and the project report itself.

Assumptions

The system assumes the availability of real time or near real time traffic data via public APIs or simulated datasets reflecting urban traffic patterns. Emergency vehicles are assumed to have GPS tracking capabilities transmitting location data to the backend server. Signal priority is assumed to be simulated, not integrated with actual municipal infrastructure due to hardware access constraints. Driver alerts assume user access to the companion application or dashboard. Connectivity is assumed stable during vehicle transit.

References

1. *Dijkstra, E. W., "A Note on Two Problems in Connexion with Graphs," Numerische Mathematik , 1959.*
2. *Hart, P. E., Nilsson, N. J., and Raphael, B., "A Formal Basis for the Heuristic Determination of Minimum Cost Paths," IEEE Transactions on Systems Science and Cybernetics , 1968.*
3. *Google Maps Platform Documentation. https://developers.google.com/maps*
4. *OpenStreetMap Documentation. https://www.openstreetmap.org*
5. *Waze for Cities Program. https://www.waze.com*
6. *Federal Highway Administration, Emergency Vehicle Preemption Guidelines .*